Rmk[section]

Josh's MAT223 Notes

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Abstract

Theorems and definitions from the second half of MAT223H1Y in Summer 2025.

10 Isomorphisms

Recall:
(1) To study linear transformations we can just study their defining matrices
(2) For a linear transformation F, im(F) is a vector subspace.
Question: How does a linear transformation change the dimension of the vector space?

10.1 Injectivity and surjectivity

Def. A function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is **injective (one-to-one)** if

$$\forall x_1, x_2 \in \mathbb{R}^n, f(x_1) = f(x_2) \implies x_1 = x_2$$

A function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is **surjective (onto)** if

$$\forall y \in \mathbb{R}^m, \exists x \in \mathbb{R}^n, \text{ s.t. } f(x) = y$$

A function $f\colon \mathbb{R}^n \to \mathbb{R}^m$ is **bijective** if it is injective and surjective.

Ex. Determine whether $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is injective, surjective, bijective, or none.

Sol. $T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$

So it is injective and surjective.

Thm 10.1. Let $f\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with defining matrix A_F.
(1) F is injective $\iff$ every column of rref(A_F) is a pivot column.
(2) F is surjective $\iff$ every row of rref(A_F) has a pivot.

Proof to be added.

10.2 Isomorphisms

Motivation: To give a formal notion of vector subspaces "looking alike". (e.g. x-z plane $\cong$ x-y plane)

Prop 10.1. Suppose $F : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation. Then if F is bijective, thenz $n = m$.

As with practice problems , can define linear transformation between any two vector spaces $V \subset \mathbb{R}^n, W \subseteq \mathbb{R}^m$.

Def. Let V be a vector subspace of $\mathbb{R}^n$ and let W be a vector subspace of $\mathbb{R}^m$. An **isomorphism** is a bijective linear transformation $F : V \to W$.

Vector spaces V and W are **isomorphic** if there is an isomorphism between V and W, which we denote by writing $V \cong W$.

Finishing isomorphisms today.

Thm 10.2. Let V and W be vector spaces. $V \cong W$ if and only if $dim(V) = dim(W)$.

It is intuitive that V and W look the same whenever they have the same dimension. It is necessary though, that they are vector spaces.

Proof. ($\Longrightarrow$) Assume $V \cong W$. Assume $dim(V) = n$ so V has a basis of n vectors $(v_1, \ldots, v_n)$. Then we claim that $(F(v_1), \ldots, F(v_n))$ is a basis of W where $F : V \to W$ is an isomorphism between V and W. Recall that this implies F is linear and bijective.

1. $(F(v_1), \ldots, F(v_n))$ is linearly independent

   Suppose for some $a_1, \ldots, a_n$ that $a_1F(v_1) + a_2F(v_2) + \cdots + a_nF(v_n) = 0$.

   By linearity, we have
   $$F(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = 0$$
   Then $a_1v_1 + \cdots + a_nv_n \in ker(F) = \{0\}$ which means that
   $$a_1v_1 + \cdots + a_nv_n = 0$$
   By linear independence of $(v_1, \ldots, v_n)$, it must be that
   $$a_1 = \cdots = a_n = 0$$

2. $(F(v_1), \ldots, F(v_n))$ spans W.

   Since F is surjective, for any $w \in W$, there exists at least one v such that $F(v) = w$.

   But $v = a_1v_1 + \cdots + a_nv_n$ for some $a_1, \ldots, a_n \in \mathbb{R}$, meaning
   $$F(v) = F(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = w$$
   By linearity of F,
   $$w = a_1F(v_1) + a_2F(v_2) + \cdots + a_nF(v_n) = 0$$
   Hence, $w \in span(F(v_1), \ldots, F(v_n))$

($\Longleftarrow$) Assume $dim(V) = dim(W) = n$. Define $F : V \to W$ by
$$F(a_1v_1 + \cdots + a_nv_n) = a_1w_1 + \cdots + a_nw_n$$
where $(v_1, v_2 \ldots, v_n)$ is a basis of V and $(w_1, w_2, \ldots w_n)$ is a basis of W. Intuitively one can see how this is linear.

Claim: F is an isomorphism. unfinished proof

$\square$

11 Inverses

11.1 Identities

Def. Let X be a set. Then the **identity** on X is the function $id_x : X \to X$ defined by $id_X(x) = x$ for any $x \in X$

Def. Let $f : X \to Y$ be a function. Then the **inverse** of f (if it exists) is the function $f^{-1} : Y \to X$ such that $f \circ f^{-1} = id_Y$ and $f^{-1} \circ f = id_X$
If such f^{-1} exists, we say f is **invertible.**

Ex. $f(x) = 3x + 4 \quad f^{-1}(y) = \frac{y-4}{3}$

Prop 11.1. If $F : \mathbb{R}^n \to \mathbb{R}^m$ is a bijective linear transformation, then it is invertible, and its inverse F^{-1} is also a linear transformation.

Proof. Since F is bijective, for any $y \in \mathbb{R}^m$ there exists a unique $x_y \in \mathbb{R}^n$ such that $F(x_y) = y$. Define $G : \mathbb{R}^m \to \mathbb{R}^n$ by $G(y) = x_y$.

Claim: (1) G is indeed an inverse of F and (2) G is a linear transformation.
(1) is intuitive by surjectivity of F. Thus it suffices to show (2).
Let $y_1, y_2 \in \mathbb{R}^n$ and $c \in \mathbb{R}$ be given arbitrarily. Want to show $G(y_1 + cy_2)$ □

11.2 Matrix Inverses

Def. The **identity matrix** I_n is the defining matrix of the identity

$$id_{\mathbb{R}^n} : \mathbb{R}^n \to \mathbb{R}^m, id_{\mathbb{R}^n}(x) = x$$

Def. (Geometric def of inverse matrix) Let A be an $n \times n$ matrix. The inverse of A (if it exists) is the defining matrix of the inverse transformation T_A^{-1}.

Def. (Algebraic def of inverse matrix) Let A be an $n \times n$ matrix. Then the inverse of A, if it exists, is the $n \times n$ matrix B such that

$$(1)AB = I_n \quad (2)BA = I_n$$

Def. (Invertibility of a matrix) We say a matrix is invertible if an inverse matrix exists.

Remarks

1. The geometric definition is equivalent to algebraic
2. $T_A^{-1} = T_{A^{-1}}$

MAT223 exists to teach us how to compute things systematically. Consider A^{-1}.

11.3 How to calculate A^{-1} systematically

Given an $n \times n$ invertible matrix A. The system of linear equations given by $A\vec{a} = \vec{b}$ has augmented matrix $\left(A \mid \vec{b}\right)$. If rref of this augmented matrix takes the form of a staircase of

1's, then we know that

$$\vec{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

Idea for computing A^{-1}: Look at multiple $\vec{a}$ corresponding to multiple $\vec{b}$ at the same time!

Assume A^{-1} takes the form $(\vec{a_1}\ \vec{a_2}\ \cdots\ \vec{a_m})$. If A^{-1} is indeed an inverse of A, we must have $AA^{-1} = I_n$, which $(A\vec{a_1} \cdots A\vec{a_n}) = (\vec{e_1}\ \vec{e_2}\ \cdots\ \vec{e_n})$

Thus we can see that $\vec{a_i}$ are the solutions of $A\vec{a_i} = \vec{e_i}$. We can solve each a_i individually:

$$(A|e_i) \rightarrow_{\text{row op}} (I_n| \begin{pmatrix} 1 \\ 2 \\ \vdots \\ n \end{pmatrix})\text{FIX THIS}$$

We can solve all the $\vec{a_i}$ at the same time:

$$(A|e_i) \rightarrow_{\text{row op}} (I_n|\vec{a_1}\ \ \vec{a_1}\ \ \vec{a_1})$$

ADD THIS PART AFTER LECTURE ENDS

Ex. Copy examples after lecture and add better matrix shortcuts

Lem 11.1. Let A be a $n \times n$ matrix. If B is a $n \times n$ matrix s.t. $AB = I_n$ then $BA = I_n$.

Proof. From the last assignment, we know that $rank(AB) \leq min(rank(A), rank(B))$. By rank nullity theorem, this implies that $nullity(AB) \geq max(nullity(A), nullity(B))$. But $nullity(AB) = nullity(I_n) = 0$ so $nullity(B) = 0$.

This then tells us that every column of B is a pivot column, every row of B is a pivot row. (B is a $n \times n$ matrix)

This implies that the matrix transformation of B has an inverse so B^{-1} exists.

Now, $A = AI_n = A(BB^{-1}) = (AB)B^{-1} = I_nB^{-1} = B^{-1}$. This implies

$$A^{-1} = B$$

□

12 Determinants in $\mathbb{R}^2$

Motivation

A linear transformation $F : \mathbb{R}^n \to \mathbb{R}^n$ can be completely determined by how it maps the standard basis:

$$F(a_1\vec{e}_1 + a_2\vec{e}_2 + \cdots + a_n\vec{e}_n) = a_1F(\vec{e}_1) + a_2F(\vec{e}_2) + \cdots + a_nF(\vec{e}_n).$$

To motivate the determinant, we observe how F acts on its domain.

12.1 Area and Orientation

Ex. Consider $F : \mathbb{R}^2 \to \mathbb{R}^2$ a linear transformation. The image $F(S)$ of the unit square S is a parallelogram with sides given by $F(\mathbf{e}_1)$ and $F(\mathbf{e}_2)$.

On the standard grid, the lengths of $F(\mathbf{e}_1)$ and $F(\mathbf{e}_2)$ determine the side lengths of $F(S)$, and the angle between them determines its shape.

Prop 12.1. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation. Then $F(S)$ is a parallelogram with sides given by $F(\mathbf{e}_1)$ and $F(\mathbf{e}_2)$. Furthermore, the standard coordinate grid for $\mathbb{R}^2$ is transformed into a grid of parallelograms aligned with $F(\mathbf{e}_1)$ and $F(\mathbf{e}_2)$.

We can gain a lot of geometric insight by looking at the area of $F(S)$.

- $\text{Area}(F(S)) > 0$ and $F(\mathbf{e}_1), F(\mathbf{e}_2)$ span $\mathbb{R}^2 \iff F$ is surjective.
- The area of $F(S)$ is the area of the parallelogram with sides $\|F(\mathbf{e}_1)\|$ and $\|F(\mathbf{e}_2)\|$.

Orientation issue: Two transformations may produce parallelograms with the same area but opposite orientation.

Def. An **ordered basis** $(\mathbf{b}_1, \mathbf{b}_2)$ for $\mathbb{R}^2$ is called *positively oriented* if we can rotate $\mathbf{b}_1$ by an angle less than $180°$ counterclockwise to obtain $\mathbf{b}_2$. Otherwise, it is called *negatively oriented*.

Ex. The basis $\{(1,0),(0,1)\}$ is positively oriented; the basis $\{(1,0),(0,-1)\}$ is negatively oriented.

12.2 Determinant of a Transformation

Def. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation. The **determinant** of F, denoted $\det(F)$, is the *oriented area* of $F(S)$:

$$\det(F) = \begin{cases} \text{Area}(F(S)), & \text{if } (F(\mathbf{e}_1), F(\mathbf{e}_2)) \text{ is positively oriented,} \\ -\text{Area}(F(S)), & \text{if } (F(\mathbf{e}_1), F(\mathbf{e}_2)) \text{ is negatively oriented.} \\ 0, & \text{if } F(\vec{e}_1), F(\vec{e}_2) \text{ are linearly dependent.} \end{cases}$$

12.3 Determinants for Matrices

Def. If A is a 2×2 matrix, the determinant of A is defined to be the determinant of its associated matrix transformation $T_A(\mathbf{x}) = A\mathbf{x}$; that is, $\det(A) = \det(T_A)$.

13 Finding Determinants on $\mathbb{R}^2$, $\mathbb{R}^3$ and Higher Dimensions

13.1 Determinants on $\mathbb{R}^2$

Def. The determinant of a 2×2 matrix A is the determinant of its matrix transform T_A, which equals the *oriented area* of the image of the unit square S under T_A, i.e.,

$$\det(A) = \mathrm{vol}(T_A(S)).$$

Ex. Let

$$A = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix}.$$

Then $T_A(e_1) = (2, 0)$, $T_A(e_2) = (2, 1)$. The oriented area of $T_A(S)$ is 2, hence $\det(A) = 2$.

Let

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then $T_B(e_1) = (1, 1)$, $T_B(e_2) = (1, -1)$. The area is 2, but the orientation is negative, so $\det(B) = -2$.

Prop 13.1. For

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

we have

$$\det(M) = ad - cb.$$

Proof. Assume $c = 0$. Then $T_M(e_1) = (a, 0)$, $T_M(e_2) = (0, d)$, so $\det(M) = ad$. For $c \neq 0$, one can transform the first column to $(a, 0)$ via column operations without changing area magnitude, leading to $\det(M) = ad - cb$. □

13.2 det on $\mathbb{R}^3$

Def. [Unit Cube] The unit cube is the subset of $\mathbb{R}^3$ given by

$$C = \{a_1 e_1 + a_2 e_2 + a_3 e_3 : 0 \leq a_1, a_2, a_3 \leq 1\}.$$

Prop 13.2. Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation. Then $F(C)$ is a parallelepiped with edges $F(e_1)$, $F(e_2)$, $F(e_3)$. The volume of $F(C)$ is

$$V(F(C)) = \det(F) \quad \text{(up to sign, depending on orientation).}$$

Def. [Orientation] An ordered basis (b_1, b_2, b_3) is *positively oriented* if it follows the **Right Hand Rule (RHR)**; otherwise it is negatively oriented.

Def. [Determinants in $\mathbb{R}^3$] Let $F : \mathbb{R}^3 \to \mathbb{R}^3$ be linear. Then

$$\det(F) = \begin{cases} V(F(C)), & \text{if } (F(e_1), F(e_2), F(e_3)) \text{ is positively oriented,} \\ -V(F(C)), & \text{if negatively oriented,} \\ 0, & \text{if } V(F(C)) = 0. \end{cases}$$

13.3 How to Compute det of a 3×3 Matrix

Prop 13.3. Given $A = (\vec{a}_1, \vec{a}_2, \vec{a}_3)$, we have:

$$\det(\vec{a}_1, \vec{a}_2, \vec{a}_3) = a_{11} \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} - a_{12} \det \begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix} + a_{13} \det \begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}.$$

Ex.

$$\det \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & 1 \\ 1 & -1 & 4 \end{pmatrix} = 1(-2) - 2((-2)(-2)) + (-1)(-4) = -2 + 4 + 4 = 6.$$

13.4 Determinants in Higher Dimensions

Def. [Minor] Let A be an $n \times n$ matrix. The (i, j)-minor A_{ij} is the $(n-1) \times (n-1)$ matrix obtained by deleting the ith row and jth column of A.

Def. [Determinants] For A an $n \times n$ matrix,

$$\det(A) = a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{n+1} a_{1n} \det(A_{1n}),$$

expanding along any row or column.

Prop 13.4. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be linear. Then

$$V(F(C_n)) = |\det(F)|,$$

where C_n is the n-dimensional unit cube.

space

14 Eigenvalues and Eigenvectors

14.1 Definition

We want to study all vectors that are stretched purely by a linear transformation T (i.e. $T\mathbf{v} = \lambda\mathbf{v}$) and their stretch factors.

Def. [Eigenvalue and Eigenvector] Let A be an $n \times n$ matrix. A nonzero vector $\mathbf{v}$ is an **eigenvector** of A if there exists a real number λ such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

If such a λ exists, we call it an **eigenvalue** of A.

*** 1.** By this definition, $\mathbf{v} \neq \mathbf{0}$ is an eigenvector with eigenvalue λ precisely when $A\mathbf{v}$ can be obtained by stretching $\mathbf{v}$ by a factor λ (possibly flipping its direction if $\lambda < 0$).

Ex. Let

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

We compute:

$$A\mathbf{v} = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} = 2 \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

(Example adapted from original notes — verify calculations as needed). If $A\mathbf{v} = 2\mathbf{v}$, then $\mathbf{v}$ is an eigenvector with eigenvalue 2.

Similarly, check if $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector: If there is no scalar λ such that $A\mathbf{v} = \lambda\mathbf{v}$, then it is not an eigenvector.

14.2 Finding Eigenvectors for a Known Eigenvalue

Prop 14.1. For an $n \times n$ matrix A, the set of eigenvectors of A corresponding to an eigenvalue λ is equal to the set of all vectors in $\text{null}(A - \lambda I_n)$, excluding $\mathbf{0}$.

Proof. If $Av = \lambda v$ with $v \neq 0$, then $(A - \lambda I_n)v = 0$, so $v \in \text{Null}(A - \lambda I_n)$ and $v \neq 0$. Conversely, if $v \in \text{Null}(A - \lambda I_n)$ with $v \neq 0$, then $(A - \lambda I_n)v = 0 \iff Av = \lambda v$, so v is an eigenvector for λ. □

Def. [Eigenspace] We call $\text{null}(A - \lambda I_n)$ the **eigenspace** of A corresponding to λ. Geometrically, E_λ is the set of vectors that A stretches (or flips) parallel to themselves by a factor λ.
The **geometric multiplicity** of λ is defined as $\dim(E_\lambda)$.

Ex. Let

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}.$$

Find E_2:

$$A - 2I = \begin{bmatrix} 1 & 5 \\ 2 & 2 \end{bmatrix}.$$

We solve $(A - 2I)\mathbf{v} = \mathbf{0}$ to find $\text{null}(A - 2I)$. Row reducing gives:

$$\begin{bmatrix} 1 & 5 \\ 0 & -8 \end{bmatrix} \Rightarrow v_2 = -\frac{1}{4}v_1.$$

Thus:

$$E_2 = \operatorname{span}\left\{\begin{bmatrix} 4 \\ -1 \end{bmatrix}\right\}.$$

14.3 Finding Eigenvalues

Prop 14.2. A scalar λ is an eigenvalue of A if and only if

$$\det(A - \lambda I_n) = 0.$$

Proof. By definition,

λ is an eigenvalue of $A \iff \exists\, v \neq 0$ such that $Av = \lambda v \iff \exists\, v \neq 0$ such that $(A - \lambda I_n)v = 0$.

Thus $A - \lambda I_n$ has a nontrivial null space, so it is not invertible. For square matrices, noninvertibility is equivalent to zero determinant; hence

$$A - \lambda I_n \text{ not invertible} \iff \det(A - \lambda I_n) = 0.$$

Combining the equivalences gives the result. □

Def. [Characteristic Polynomial] For an $n \times n$ matrix A, the **characteristic polynomial** of A is

$$\chi_A(\lambda) = \det(A - \lambda I_n).$$

*** 2.** The eigenvalues of A are precisely the roots of $\chi_A(\lambda)$.

Ex. Let

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}.$$

The characteristic polynomial is:

$$\chi_A(\lambda) = \det \begin{bmatrix} 3-\lambda & 5 \\ 2 & 4-\lambda \end{bmatrix} = (3-\lambda)(4-\lambda) - 10 = \lambda^2 - 7\lambda + 2.$$

Solving $\chi_A(\lambda) = 0$ gives the eigenvalues.

14.4 Summary: Strategy for Finding All Eigenvalues and Eigenvectors

Given an $n \times n$ matrix A:

1. Find all eigenvalues λ of A by solving $\chi_A(\lambda) = 0$.
2. For each eigenvalue λ, find the eigenspace $E_\lambda = \operatorname{null}(A - \lambda I_n)$.
3. The nonzero vectors in E_λ are the eigenvectors corresponding to λ.

15 Finding Eigenvalues and Eigenvectors; Coordinate Systems and Change of Basis

15.1 Eigenvalues and Eigenvectors

Recall:

Def. (Summary of solving eigenvalues and eigenvectors)
1. Let T be a linear transformation. An *eigenvector* of T with eigenvalue λ is a nonzero vector $\vec{v}$ such that $T(\vec{v}) = \lambda\vec{v}$.
2. Given a matrix A, the eigenvalues of A are found by solving

$$\chi_A(\lambda) = \det(A - \lambda I_n) = 0.$$

For each eigenvalue λ, the corresponding eigenspace is

$$E_\lambda = \text{null}(A - \lambda I_n).$$

Ex. Given

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

the characteristic polynomial is

$$\chi_A(\lambda) = (1-\lambda)^2(2-\lambda).$$

Thus $\lambda = 1$ (algebraic multiplicity 2) and $\lambda = 2$ (algebraic multiplicity 1).

For $\lambda = 1$:

$$\text{null}(A - I_3) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, \quad \dim = 2.$$

For $\lambda = 2$:

$$\text{null}(A - 2I_3) = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \dim = 1.$$

*** 3.** The algebraic multiplicity of an eigenvalue is not necessarily equal to its geometric multiplicity.

15.2 Coordinate Systems and Change of Basis

Def. [Coordinates given by a basis] Let $B = (\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n)$ be an ordered basis of a vector space V. Any $\vec{x} \in V$ can be uniquely expressed as

$$\vec{x} = x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n.$$

The B-coordinates of $\vec{x}$ are given by

$$[\vec{x}]_B = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Ex. Let $\vec{b}_1 = (1,1,0)$, $\vec{b}_2 = (2,-1,0)$, $\vec{b}_3 = (3,4,1)$, and $\vec{x} = (1,0,2)$. To find $[\vec{x}]_B$, solve $x_1\vec{b}_1 + x_2\vec{b}_2 + x_3\vec{b}_3 = \vec{x}$ by row-reducing the augmented matrix

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & -1 & 4 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

to obtain $x_1 = 3$, $x_2 = -1$, $x_3 = 2$.

Thm 15.1. Let V be a subspace of $\mathbb{R}^n$ and B a basis of V. Every $\vec{x} \in V$ has a unique representation in terms of B:

$$\vec{x} = x_1\vec{b}_1 + x_2\vec{b}_2 + \cdots + x_n\vec{b}_n.$$

Proof. (*Existence.*) Since B is a basis of V, it spans V. Hence for any $\mathbf{x} \in V$ there exist scalars $x_1, \ldots, x_k$ such that $\mathbf{x} = \sum_{i=1}^{k} x_i\,\mathbf{b}_i$.

(*Uniqueness.*) Suppose $\mathbf{x} \in V$ has two representations with respect to B:

$$\mathbf{x} = \sum_{i=1}^{k} x_i\,\mathbf{b}_i = \sum_{i=1}^{k} y_i\,\mathbf{b}_i.$$

Subtracting,

$$\mathbf{0} = \sum_{i=1}^{k} (x_i - y_i)\,\mathbf{b}_i.$$

Because B is a basis, the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_k$ are linearly independent; therefore all coefficients must be zero:

$$x_i - y_i = 0 \quad \text{for each } i = 1, \ldots, k.$$

Hence $x_i = y_i$ for all i, so the representation is unique. □

Def. [Change of basis matrix] Let C and B be bases for V. The change of basis matrix $M_{C\leftarrow B}$ satisfies

$$[\vec{x}]_C = M_{C\leftarrow B}\,[\vec{x}]_B, \quad \forall \vec{x} \in V.$$

Ex. Let $B = (\vec{b}_1, \vec{b}_2)$, $C = (\vec{c}_1, \vec{c}_2)$, with $\vec{c}_1 = (1,0)$, $\vec{c}_2 = (5,3)$. If $\vec{b}_1 = (2,1)$ and $\vec{b}_2 = (5,3)$, then

$$M_{C\leftarrow B} = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}.$$

Similarly, $M_{B\leftarrow C} = (M_{C\leftarrow B})^{-1}$.

16 Change of Basis Matrix

16.1 Recall

If $B = (\vec{b}_1, \dots, \vec{b}_n)$ is a basis of $\mathbb{R}^n$ and Σ is the standard basis, then

$$M_{\Sigma \leftarrow B} = \big([\vec{b}_1]_\Sigma \; [\vec{b}_2]_\Sigma \; \cdots \; [\vec{b}_n]_\Sigma\big), \qquad M_{B \leftarrow \Sigma} = (M_{\Sigma \leftarrow B})^{-1}.$$

Lem 16.1. Let $\mathcal{C}$ be a basis of a finite-dimensional vector space V. Then the coordinate map $x \mapsto [x]_\mathcal{C}$ is linear: for all $\vec{x}, \vec{y} \in V$ and $k \in \mathbb{R}$,

$$[\vec{x} + \vec{y}]_\mathcal{C} = [\vec{x}]_\mathcal{C} + [\vec{y}]_\mathcal{C}, \qquad [k\vec{x}]_\mathcal{C} = k[\vec{x}]_\mathcal{C}.$$

Thm 16.1. Let $B = (\vec{b}_1, \dots, \vec{b}_n)$ be a basis of $\mathbb{R}^n$ and let Σ be the standard basis. For any $\vec{x} \in \mathbb{R}^n$,

$$[\vec{x}]_\Sigma = M_{\Sigma \leftarrow B} \, [\vec{x}]_B \quad \text{with} \quad M_{\Sigma \leftarrow B} = \big([\vec{b}_1]_\Sigma \; [\vec{b}_2]_\Sigma \; \cdots \; [\vec{b}_n]_\Sigma\big).$$

Moreover, $M_{\Sigma \leftarrow B}$ is invertible and

$$M_{B \leftarrow \Sigma} = (M_{\Sigma \leftarrow B})^{-1}.$$

Proof. Write $[\vec{x}]_B = (x_1, \dots, x_n)^\mathsf{T}$; then $\vec{x} = x_1\vec{b}_1 + \cdots + x_n\vec{b}_n$. By Lemma 16.1,

$$[\vec{x}]_\Sigma = x_1[\vec{b}_1]_\Sigma + \cdots + x_n[\vec{b}_n]_\Sigma = \big([\vec{b}_1]_\Sigma \; \cdots \; [\vec{b}_n]_\Sigma\big)[\vec{x}]_B.$$

Since B is a basis, the columns $[\vec{b}_i]_\Sigma$ are linearly independent, hence $M_{\Sigma \leftarrow B}$ is invertible. Multiplying the displayed equation by $(M_{\Sigma \leftarrow B})^{-1}$ yields $[\vec{x}]_B = (M_{\Sigma \leftarrow B})^{-1}[\vec{x}]_\Sigma$, i.e. $M_{B \leftarrow \Sigma} = (M_{\Sigma \leftarrow B})^{-1}$. □

Lem 16.2. Let V be a vector space with basis $B = (\vec{b}_1, \dots, \vec{b}_n)$. For any basis $\mathcal{C}$ of V, the list $\big([\vec{b}_1]_\mathcal{C}, \dots, [\vec{b}_n]_\mathcal{C}\big)$ is linearly independent in $\mathbb{R}^n$.

Proof. Let $a_1, \dots, a_n \in \mathbb{R}$ with $\sum_i a_i[\vec{b}_i]_\mathcal{C} = \vec{0}$. By Lemma 16.1, $\big[\sum_i a_i\vec{b}_i\big]_\mathcal{C} = \sum_i a_i[\vec{b}_i]_\mathcal{C} = \vec{0}$, so $\sum_i a_i\vec{b}_i = \vec{0}$. Since B is a basis, all $a_i = 0$. □

Thm 16.2. Let V be a vector space with basis $\mathcal{C} = (\vec{c}_1, \dots, \vec{c}_n)$. A list $B = (\vec{b}_1, \dots, \vec{b}_n)$ in V is a basis of V iff the matrix

$$M_{\mathcal{C} \leftarrow B} = \big([\vec{b}_1]_\mathcal{C} \; [\vec{b}_2]_\mathcal{C} \; \cdots \; [\vec{b}_n]_\mathcal{C}\big)$$

is invertible. Also, notice that this means the change of basis matrix exists. In this case,

$$M_{B \leftarrow \mathcal{C}} = (M_{\mathcal{C} \leftarrow B})^{-1}.$$

Proof. ($\Rightarrow$) If B is a basis, then as in the proof of Theorem 16.1, $[\vec{x}]_\mathcal{C} = M_{\mathcal{C} \leftarrow B}[\vec{x}]_B$ for all $\vec{x} \in V$, and by Lemma 16.2 the columns $[\vec{b}_i]_\mathcal{C}$ are linearly independent, so $M_{\mathcal{C} \leftarrow B}$ is invertible. Hence $M_{B \leftarrow \mathcal{C}} = (M_{\mathcal{C} \leftarrow B})^{-1}$.

($\Leftarrow$) If $M_{\mathcal{C} \leftarrow B}$ is invertible, its columns are linearly independent, so if $\sum_i a_i\vec{b}_i = \vec{0}$, then applying $[\,\cdot\,]_\mathcal{C}$ and using Lemma 16.1 gives $\sum_i a_i[\vec{b}_i]_\mathcal{C} = \vec{0}$, forcing all $a_i = 0$. Thus B is linearly independent. Since $\dim V = n$, any n linearly independent vectors span V, so B is a basis. □

Ex. Let $\mathcal{C} = (\vec{c}_1, \vec{c}_2)$ with $\vec{c}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\vec{c}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, a basis of a 2-dimensional subspace $V \subset \mathbb{R}^2$. Let $B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} \frac{4}{3} \\ \frac{2}{3} \end{pmatrix} \right)$. Then

$$\left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} \right]_{\mathcal{C}} = \left[\vec{c}_1 + \vec{c}_2 \right]_{\mathcal{C}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \left[\begin{pmatrix} \frac{4}{3} \\ \frac{2}{3} \end{pmatrix} \right]_{\mathcal{C}} = \left[2\vec{c}_1 + \vec{c}_2 \right]_{\mathcal{C}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Hence

$$M_{\mathcal{C} \leftarrow B} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad \det M_{\mathcal{C} \leftarrow B} = -1 \neq 0,$$

so B is also a basis of V by Theorem 16.2. Moreover,

$$M_{B \leftarrow \mathcal{C}} = (M_{\mathcal{C} \leftarrow B})^{-1} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}.$$

17 Similarity and Diagonalization

17.1 Defining Matrix of a Linear Transformation w.r.t. a Basis

Recall: If $F : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation and Σ is the standard basis, then the **defining matrix** of F with respect to Σ is

$$A_{F,\Sigma} = \big([F(\vec{e}_1)]_\Sigma \ [F(\vec{e}_2)]_\Sigma \ \cdots \ [F(\vec{e}_n)]_\Sigma\big),$$

and $[F(\vec{x})]_\Sigma = A_{F,\Sigma}[\vec{x}]_\Sigma$ for all $\vec{x} \in \mathbb{R}^n$.

Thm 17.1. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be linear, and let $B = (\vec{b}_1, \ldots, \vec{b}_n)$ be any basis of $\mathbb{R}^n$. There exists a unique $n \times n$ matrix A such that

$$[F(\vec{x})]_B = A[\vec{x}]_B$$

for all $\vec{x}$, and

$$A = \big([F(\vec{b}_1)]_B \ [F(\vec{b}_2)]_B \ \cdots \ [F(\vec{b}_n)]_B\big).$$

Proof. Let $\vec{x} = x_1\vec{b}_1 + \cdots + x_n\vec{b}_n$. Then

$$[F(\vec{x})]_B = x_1[F(\vec{b}_1)]_B + \cdots + x_n[F(\vec{b}_n)]_B = \begin{bmatrix}[F(\vec{b}_1)]_B & \cdots & [F(\vec{b}_n)]_B\end{bmatrix}\begin{bmatrix}x_1\\ \vdots \\ x_n\end{bmatrix}.$$

□

Def. [Defining matrix w.r.t. a basis] Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be linear, and let B be a basis of $\mathbb{R}^n$. The defining matrix of F with respect to B is

$$A_{F,B} = \big([F(\vec{b}_1)]_B \ [F(\vec{b}_2)]_B \ \cdots \ [F(\vec{b}_n)]_B\big),$$

where $[F(\vec{x})]_B = A_{F,B}[\vec{x}]_B$ for all $\vec{x}$.

Ex. Let $B = (\vec{b}_1, \vec{b}_2)$ with $\vec{b}_1 = \begin{pmatrix}1\\1\end{pmatrix}$ and $\vec{b}_2 = \begin{pmatrix}-2\\1\end{pmatrix}$. Let $F(x, y) = (2x + y, x - y)$. Then

$$A_{F,\Sigma} = \begin{pmatrix}2 & 1\\ 1 & -1\end{pmatrix}.$$

To find $A_{F,B}$, we compute $[F(\vec{b}_1)]_B$ and $[F(\vec{b}_2)]_B$:

$$F(\vec{b}_1) = (3, 0) = -1\vec{b}_1 + \vec{b}_2 \quad \Rightarrow \quad [F(\vec{b}_1)]_B = \begin{pmatrix}-1\\1\end{pmatrix},$$

$$F(\vec{b}_2) = (-3, 0) = -3\vec{b}_1 + 0\vec{b}_2 \quad \Rightarrow \quad [F(\vec{b}_2)]_B = \begin{pmatrix}-3\\0\end{pmatrix}.$$

Thus

$$A_{F,B} = \begin{pmatrix}-1 & -3\\ 1 & 0\end{pmatrix}.$$

17.2 Matrix similarity

Def. [Geometric definition] Two $n \times n$ matrices B and C are *similar* if they represent the same linear transformation in possibly different bases. That is, there exists a basis B with $A_{F,B} = B$ and a basis C with $A_{F,C} = C$ for the same F.

Def. [Algebraic definition] Two $n \times n$ matrices B and C are similar if there exists an invertible P such that

$$B = P^{-1}CP.$$

If B and C are similar, B is invertible iff C is invertible.

Thm 17.2. If B and C are similar, then B is invertible $\iff$ C is invertible.

Ex. From the previous example:

$$\begin{pmatrix} -1 & -3 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

are similar.

17.3 Diagonalization

Def. [Diagonal matrix] A matrix is *diagonal* if all nonzero entries lie on the main diagonal. We write

$$D = \operatorname{diag}(d_1, \ldots, d_n).$$

Def. [Diagonalizable] An $n \times n$ matrix A is *diagonalizable* if it is similar to a diagonal matrix.

Thm 17.3 (Diagonalization theorem)**.** A matrix A is diagonalizable iff A has n linearly independent eigenvectors $\vec{v}_1, \ldots, \vec{v}_n$. In this case, letting $C = (\vec{v}_1 \ \cdots \ \vec{v}_n)$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $A\vec{v}_i = \lambda_i \vec{v}_i$, we have

$$D = C^{-1}AC.$$

Proof. ($\Rightarrow$) If A is diagonalizable, $A = CDC^{-1}$ with C invertible. Columns of C are eigenvectors. ($\Leftarrow$) If A has n independent eigenvectors, let B be this eigenbasis. Then A has a diagonal defining matrix in basis B, hence is diagonalizable. $\square$

Ex. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$. The characteristic polynomial is

$$\chi_A(\lambda) = \det(A - \lambda I) = (\lambda - 4)(\lambda + 1),$$

so $\lambda_1 = 4$ and $\lambda_2 = -1$.

An eigenvector for $\lambda_1 = 4$ is $\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and for $\lambda_2 = -1$ is $\vec{v}_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

Then $B = (\vec{v}_1, \vec{v}_2)$ is a basis of eigenvectors, and

$$D = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad A = M_{\Sigma \leftarrow B} D M_{B \leftarrow \Sigma}.$$

18 Wrapping up the Diagonalization Theorem

Recall

Thm 18.1 (Diagonalization Theorem)**.** If A is an $n \times n$ matrix, then A is diagonalizable iff A has n linearly independent eigenvectors $\vec{v}_1, \ldots, \vec{v}_n$. In this case, $D = C^{-1}AC$ with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $C = (\vec{v}_1 \ \cdots \ \vec{v}_n)$.

18.1 Facts

Prop 18.1. Eigenvectors of distinct eigenvalues of a matrix are linearly independent.

Ex. If A has eigenvalues 1 and 2 and $\vec{v}_1, \vec{v}_2$ are eigenvectors for 1 and 2, then $(\vec{v}_1, \vec{v}_2)$ is linearly independent.

Cor 18.1. If A has n distinct eigenvalues, then A is diagonalizable.

Proof. Having n distinct eigenvalues implies the existence of n linearly independent eigenvectors. By the diagonalization theorem, A is diagonalizable. □

*** 4.** The converse is false: A may be diagonalizable without having n distinct eigenvalues.

Ex. $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable, has eigenvalues $1, 2$ with $n = 3$.

18.2 Characterizing diagonalizability with repeated eigenvalues

Thm 18.2 (Diagonalization Theorem, final version)**.** Let A be an $n \times n$ matrix with all real eigenvalues. The following are equivalent:

1. A is diagonalizable.
2. The sum of the geometric multiplicities of the eigenvalues of A equals n.
3. For each eigenvalue λ of A, the geometric multiplicity equals the algebraic multiplicity.

Def. The *geometric multiplicity* of λ is $\dim(E_\lambda)$. The *algebraic multiplicity* of λ is the exponent of $(\lambda - x)$ in the characteristic polynomial $\chi_A(x) = \det(A - xI)$.

Lem 18.1. If λ is an eigenvalue of A, then its geometric multiplicity is $\leq$ its algebraic multiplicity.

Idea. Let λ be an eigenvalue with geometric multiplicity d, and let $\{\vec{v}_1, \ldots, \vec{v}_d\}$ be a basis of E_λ. Extend to a basis B of $\mathbb{R}^n$. In the basis B, the defining matrix of $F = T_A$ has the block form:

$$A_{F,B} = \begin{pmatrix} \lambda I_d & * \\ 0 & Y \end{pmatrix},$$

so

$$\chi_A(x) = \chi_{A_{F,B}}(x) = (\lambda - x)^d \cdot g(x),$$

implying algebraic multiplicity $\geq d$. □

*** 5.** To show A is not diagonalizable, it suffices to find an eigenvalue whose geometric multiplicity is less than its algebraic multiplicity.

Ex. $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 5 & 4 \end{pmatrix}$ is not diagonalizable.

Example: Non-diagonalizable matrix

Let

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -5 & 4 \end{pmatrix}.$$

The characteristic polynomial is

$$\chi_A(\lambda) = -(\lambda - 2)(\lambda - 1)^2,$$

so the algebraic multiplicity of $\lambda = 1$ is 2. Solving $(A - I_3)\vec{x} = \vec{0}$ shows $\dim E_1 = 1$, so the geometric multiplicity < 2 and A is not diagonalizable.

19 Orthogonality

19.1 Eigen-decomposition

Def. [Eigen-decomposition] If A is an $n \times n$ diagonalizable matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$ and n linearly independent eigenvectors $\vec{v}_1, \ldots, \vec{v}_n$, then

$$A = CDC^{-1}, \quad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \quad C = (\vec{v}_1 \ \cdots \ \vec{v}_n)$$

is called the *eigen-decomposition* of A.

Ex. Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ stretch vectors in the x-direction by 2 and in the y-direction by 3. Then

$$D = \operatorname{diag}(2, 3), \quad C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A_F = CDC^{-1} = \operatorname{diag}(2, 3).$$

19.2 Orthogonality and dot product

Def. [Dot product] For $\vec{u}, \vec{v} \in \mathbb{R}^n$,

$$\vec{u} \cdot \vec{v} = u_1 v_1 + \cdots + u_n v_n = \vec{u}^T \vec{v}.$$

Prop 19.1. For $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

1. Commutativity: $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$.
2. Distributivity: $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$.
3. Scalar multiplication: $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v})$.

Def. [Norm] The *norm* of $\vec{v}$ is $\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}}$.

Def. [Distance] The distance between $\vec{u}$ and $\vec{v}$ is $d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$.

Def. [Angle between vectors] If $\vec{u}, \vec{v}$ are non-parallel in $\mathbb{R}^n$, the angle θ between them satisfies

$$\cos\theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|}.$$

If $\vec{u} \cdot \vec{v} = 0$, they are *orthogonal*.

19.3 Orthogonal bases and orthogonal matrices

Def. [Orthogonal basis] A basis $(\vec{v}_1, \ldots, \vec{v}_n)$ is orthogonal if $\vec{v}_i \cdot \vec{v}_j = 0$ for $i \neq j$. If also $\|\vec{v}_i\| = 1$ for all i, it is *orthonormal*.

Prop 19.2. Let B be an orthonormal basis for $\mathbb{R}^n$. Then for any $\vec{x}, \vec{y} \in \mathbb{R}^n$:

$$[\vec{x}]_B \cdot [\vec{y}]_B = \vec{x} \cdot \vec{y}$$

In particular, $\|[\vec{x}]_B\| = \|\vec{x}\|$.

Proof. Assume $B = (\vec{v}_1, \ldots, \vec{v}_n)$ is an orthonormal basis. This means that:

$$\vec{v}_i \cdot \vec{v}_j = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$ be such that:

$$\vec{x} = x_1\vec{v}_1 + \cdots + x_n\vec{v}_n, \quad \vec{y} = y_1\vec{v}_1 + \cdots + y_n\vec{v}_n$$

Then:

$$\vec{x} \cdot \vec{y} = (x_1\vec{v}_1 + \cdots + x_n\vec{v}_n) \cdot (y_1\vec{v}_1 + \cdots + y_n\vec{v}_n)$$

By distributivity:

$$= \sum_{i,j=1}^{n} x_i y_j (\vec{v}_i \cdot \vec{v}_j)$$

Since $\vec{v}_i \cdot \vec{v}_j = 1$ when $i = j$, this becomes:

$$= \sum_{i=1}^{n} x_i y_i$$

$$= [\vec{x}]_B \cdot [\vec{y}]_B$$

□

Def. [Orthogonal matrix] A matrix Q is orthogonal if $Q^{-1} = Q^T$, equivalently if its columns form an orthonormal basis.

Prop 19.3. If Q is an $n \times n$ orthogonal matrix, then for any $\vec{v}, \vec{w} \in \mathbb{R}^n$:

$$(Q\vec{v}) \cdot (Q\vec{w}) = \vec{v} \cdot \vec{w}$$

In particular, $\|Q\vec{v}\| = \|\vec{v}\|$ and the angle between $Q\vec{v}$ and $Q\vec{w}$ equals the angle between $\vec{v}$ and $\vec{w}$.

Proof. Since Q is orthogonal, $Q^{-1} = Q^T$. Then:

$$(Q\vec{v}) \cdot (Q\vec{w}) = (Q\vec{v})^T (Q\vec{w})$$

By the transpose property:

$$= \vec{v}^T Q^T Q \vec{w}$$

Since $Q^T Q = I$:

$$= \vec{v}^T \vec{w}$$

$$= \vec{v} \cdot \vec{w}$$

□

20 The Gram–Schmidt Process, Spectrum Theorem, Singular Value Decomposition

20.1 The Gram–Schmidt Process

Motivation

In $\mathbb{R}^2$, given $\vec{V}$ and $\vec{W}$, we want $\vec{V}_\perp$ such that $\vec{V}_\parallel \parallel \vec{W}$ and $\vec{V}_\perp \perp \vec{W}$.

If $\vec{V}_\parallel = C\vec{W}$, then since $\vec{V}_\perp \cdot \vec{W} = 0$:

$$(\vec{V} - \vec{V}_\parallel) \cdot \vec{W} = 0$$

$$\vec{V} \cdot \vec{W} - C\vec{W} \cdot \vec{W} = 0$$

$$C = \frac{\vec{V} \cdot \vec{W}}{\vec{W} \cdot \vec{W}}, \quad \vec{V}_\parallel = \frac{\vec{V} \cdot \vec{W}}{\vec{W} \cdot \vec{W}}\vec{W}, \quad \vec{V}_\perp = \vec{V} - \vec{V}_\parallel$$

Def. [Orthogonal Projection] For $\vec{v}, \vec{w} \in \mathbb{R}^n$, the orthogonal projection of $\vec{v}$ onto $\vec{w}$ is:

$$\text{proj}_{\vec{w}}\vec{v} = \frac{\vec{v} \cdot \vec{w}}{\vec{w} \cdot \vec{w}}\vec{w}$$

This is also the closest point on $\text{span}(\vec{w})$ to $\vec{v}$.

Thm 20.1 (Gram–Schmidt Process)**.** Every finite-dimensional vector space has a basis of orthogonal vectors. If V is a subspace of $\mathbb{R}^n$ with basis $(\vec{v}_1, \ldots, \vec{v}_n)$, define:

$$\vec{u}_1 = \vec{v}_1$$

$$\vec{u}_2 = \vec{v}_2 - \text{proj}_{\vec{u}_1}\vec{v}_2$$

$$\vec{u}_3 = \vec{v}_3 - \text{proj}_{\vec{u}_1}\vec{v}_3 - \text{proj}_{\vec{u}_2}\vec{v}_3$$

$$\vdots$$

$$\vec{u}_n = \vec{v}_n - \sum_{k=1}^{n-1} \text{proj}_{\vec{u}_k}\vec{v}_n$$

Then $(\vec{u}_1, \ldots, \vec{u}_n)$ is an orthogonal basis.

Ex. Let $V = \text{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$ with:

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \quad \vec{v}_3 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

Applying Gram–Schmidt yields an orthogonal list $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$.

*** 6.** An orthogonal list of vectors is linearly independent. Normalizing $\vec{u}_i$ yields an **orthonormal basis**.

20.2 Spectrum Theorem

Def. A matrix A is **orthogonally diagonalizable** if there exists an orthogonal Q and

diagonal D such that:

$$Q^T A Q = D$$

Thm 20.2 (Spectrum Theorem)**.** A matrix A is orthogonally diagonalizable $\iff$ A is symmetric ($A^T = A$).

Lem 20.1. Let A be an $n \times n$ symmetric matrix. Then:

1. A has at least one real eigenvalue.
2. Eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof of Lemma. (1) By the Fundamental Theorem of Algebra, A has an eigenvalue λ with eigenvector $\vec{v} \neq \vec{0}$. Symmetry implies λ is real.

(2) If $\lambda \neq \mu$ are distinct eigenvalues with $\vec{v}, \vec{w}$ eigenvectors, symmetry implies:

$$(\lambda - \mu)(\vec{v} \cdot \vec{w}) = 0 \quad \Rightarrow \quad \vec{v} \cdot \vec{w} = 0$$

□

Proof of Theorem. ($\Rightarrow$) If A is orthogonally diagonalizable, $A = QDQ^T$ with D diagonal and Q orthogonal, then $A^T = A$.

($\Leftarrow$) If A is symmetric, the lemma ensures an orthonormal basis of eigenvectors exists, giving $A = QDQ^T$. □

20.3 Singular Value Decomposition

Prop 20.1. Let A be an $m \times n$ matrix. There exists an orthonormal basis of $\mathbb{R}^n$ of eigenvectors of $A^T A$, $(\vec{v}_1, \ldots, \vec{v}_n)$. If $A\vec{v}_1, \ldots, A\vec{v}_r$ are nonzero, these form an orthogonal basis for $\mathrm{col}(A)$.

Proof. $A^T A$ is symmetric, hence orthogonally diagonalizable. Eigenvectors $\vec{v}_i$ of $A^T A$ with eigenvalues σ_i^2 give orthogonal vectors $A\vec{v}_i$. □

21 Singular Value Decomposition

Motivation. We want to show that *any* matrix can be expressed by a rotation/reflection, followed by a dilation, and then a rotation/reflection, no matter whether A is symmetric or not.

Prop 21.1. Let A be an $m \times n$ matrix. Then there exists an orthonormal basis $(v_1, \ldots, v_n)$ of $\mathbb{R}^n$ of eigenvectors of $A^\top A$ so that

$$(Av_1, Av_2, \ldots, Av_n) \quad \text{is an orthogonal subset of } \mathbb{R}^m.$$

Furthermore, after re–indexing assume $Av_1, \ldots, Av_r \neq 0$ and $Av_{r+1} = \cdots = Av_n = 0$. Then

$$(Av_1, \ldots, Av_r) \text{ form an orthogonal basis for } \operatorname{col}(A).$$

Proof. $A^\top A$ is symmetric, hence has an orthonormal eigenbasis (v_i). For $i \neq j$, $(Av_i)^\top (Av_j) = v_i^\top A^\top A v_j = \lambda_j v_i^\top v_j = 0$ so (Av_i) are mutually orthogonal. If $y \in \operatorname{col}(A)$, write $y = Ax$ and expand $x = \sum_i \alpha_i v_i$; then $Ax = \sum_{i:\lambda_i > 0} \alpha_i A v_i$, so the nonzero Av_i span $\operatorname{col}(A)$ and are orthogonal. □

Ex. Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & -1 \end{bmatrix}$. Then

$$A^\top A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \quad \text{(symmetric)}.$$

The characteristic polynomial is $\det(A^\top A - \lambda I) = (3-\lambda)(6-\lambda) - 4 = \lambda^2 - 9\lambda + 14 = (\lambda - 7)(\lambda - 2)$, so $\lambda_1 = 7,\ \lambda_2 = 2$. An orthonormal eigenbasis is

$$v_1 = \tfrac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad v_2 = \tfrac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Then

$$Av_1 = \tfrac{1}{\sqrt{5}} \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}, \quad Av_2 = \tfrac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 0 \\ -3 \end{bmatrix},$$

which are orthogonal with $\|Av_1\| = \sqrt{7},\ \|Av_2\| = \sqrt{2}$. Let $Q = (v_1\ v_2)$, let $\Sigma = \begin{bmatrix} \sqrt{7} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}$ (3×2), and let U be a 3×3 orthogonal matrix whose first two columns are $Av_1/\sqrt{7}$ and $Av_2/\sqrt{2}$. Then $A = U \Sigma Q^\top$.

Thm 21.1 (Singular Value Decomposition)**.** Let A be an $m \times n$ matrix. Then there exists an orthogonal $m \times m$ matrix U, an orthogonal $n \times n$ matrix Q, and a "block diagonal" matrix Σ such that

$$A = U\,\Sigma\,Q^\top,$$

where Σ is $m \times n$ with diagonal entries $\sigma_1 \geq \cdots \geq \sigma_k > 0$ (and zeros elsewhere). In particular:

$$AQ = \big(Av_1, \ldots, Av_n\big) = \big(\sigma_1 u_1, \ldots, \sigma_n u_n\big) = U\Sigma.$$

Proof. Let $(v_1, \ldots, v_n)$ be an orthonormal eigenbasis of $A^\top A$, and set

$$\sigma_i = \|Av_i\|, \quad u_i = \begin{cases} Av_i/\sigma_i, & \sigma_i > 0, \\ 0, & \sigma_i = 0. \end{cases}$$

By the proposition, the nonzero u_i are orthonormal in $\mathrm{col}(A)$; extend them to an orthonormal basis of $\mathbb{R}^m$ to form U. Let $Q = (v_1 \ \cdots \ v_n)$ and let Σ place the σ_i on the (rectangular) diagonal. Then $AQ = U\Sigma$, hence $A = U\Sigma Q^\top$. $\square$

Prop 21.2. Let A be $m \times n$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of $A^\top A$. Then $\lambda_i \geq 0$, and the singular values of A are $\sigma_i = \sqrt{\lambda_i}$.

Def. [Singular values] If A is $m \times n$ and $(v_1, \ldots, v_n)$ is an orthonormal basis of $\mathbb{R}^n$ consisting of the eigenvectors of $A^\top A$, the **singular values** of A are $\sigma_i = \|Av_i\|$.

Proof.

$$\sigma_i^2 = \|Av_i\|^2 = (Av_i)^\top Av_i = v_i^\top A^\top Av_i = \lambda_i \, v_i^\top v_i = \lambda_i \|v_i\|^2 = \lambda_i,$$

so $\lambda_i \geq 0$ and $\sigma_i = \sqrt{\lambda_i}$. $\square$